

Digital Egypt Pioneers Initiative

Round 4
2025 - 2026

رواد مصر الرقمية

الدورة الرابعة
2025 - 2026

Track: AI & Data Science - Microsoft Machine Learning Engineer

Group Code: ONL4_AIS2_S7

Instructor: Eman Elgalad

Company Name: Skills Dynamix



Project Name

"AI-Powered Predictive Maintenance for Industrial Equipment"

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Testing & Quality Assurance

Test Cases & Test Plan:

- **Authentication Testing**

- Verify user registration, login, password validation, and role-based access control.

- **Data Loading & Preprocessing Testing**

- Ensure NASA CMAPSS datasets are loaded correctly and preprocessing pipelines handle inputs properly.

- **Prediction Testing**

- Validate RUL prediction functionality and verify model outputs are generated correctly.

- **Explainable AI Testing**

- Ensure SHAP explanations are generated and correctly identify important features affecting predictions.

- **RAG Assistant Testing**

- Verify knowledge retrieval, context generation, and AI responses based on the maintenance knowledge base.

- **Dashboard Testing**

- Validate Streamlit pages, visualizations, navigation, and user interactions.

- **Database Testing**

- Ensure user records, prediction history, and maintenance records are stored and retrieved correctly.

Automated Testing:

- **Testing includes**
 - Unit testing for core services and functions.
 - Validation of input/output behavior for prediction APIs.
 - Testing database operations and application workflows.
 - Checking model loading and inference functionality.
- **Automated tests are implemented to improve reliability and reduce manual verification effort.**

Bug Reports & Resolution:

- **Issues & Solutions**

- Issue: Incorrect dataset loading paths.
- Solution: Improved path handling and validation.
- Issue: Model prediction errors.
- Solution: Added input validation and preprocessing checks.
- Issue: Authentication failures.
- Solution: Fixed database integration and password validation.
- Issue: Dashboard integration issues.
- Solution: Improved communication between frontend and backend services.
- Issue: Missing dependencies.
- Solution: Updated requirements and environment configuration.